

Dr. Doc:

## OVERVIEW OF THE RESPIRATORY SYSTEM

▼ The respiratory system is structurally and functionally divided into two major components, characterized by continuous, gradual transitions in epithelia and supporting tissues rather than abrupt changes.

▼ The entire tract represents a series of pliable tubes lined by respiratory mucosa, with walls stabilized by variable amounts of smooth muscle, elastic fibers, and cartilage.

### Functional Divisions

▼ **Conducting Portion:** Serves to deliver, filter, warm, and humidify inspired air. It extends from the nasal cavity through the pharynx, larynx, trachea, bronchi, and bronchioles, terminating at the terminal bronchioles. No gas exchange occurs in this portion.

▼ **Respiratory Portion:** The site of gas exchange between inhaled air and blood. It begins where alveoli interrupt the airway walls, extending from the respiratory bronchioles to the alveolar ducts, alveolar sacs, and alveoli inclusive.

### General Wall Layers of the Conducting Portion

The walls of conducting passages (most prominent in the trachea and extra-pulmonary bronchi) typically display five distinct histological layers:

#### 1. Respiratory Epithelium

▼ A specialized lining that acts as a physical and immunological barrier.

#### 2. Lamina Propria:

- ▼ Subepithelial loose fibroelastic connective tissue.

- ▼ It is highly vascularized with thin-walled, wide-lumen superficial blood vessels specialized for warming incoming air, and rich in immune cells (lymphocytes, plasma cells, mast cells).

- ▼ It houses Bronchus-Associated Lymphoid Tissue (BALT), a subset of MALT where lymphocytes produce secretory IgA to neutralize viruses and bacteria.

### 3. Muscle Layer

- ▼ A layer of smooth muscle lying deep to the mucosa (present throughout the conducting tract, with the notable exception of the trachea).

### 4. Submucosa

- ▼ Present in extra-pulmonary airways only. It consists of connective tissue housing mixed sero-mucous glands whose watery and mucus secretions moisten, filter, and detoxify soluble gases.

### 5. Adventitia

- ▼ The outermost loose connective tissue layer that blends into surrounding lung parenchyma.

- ▼ In larger airways, it contains hyaline cartilage rings or plates for structural support.

- ▼ As the respiratory tree branches and decreases in diameter, the wall structure simplifies.

- ▼ The epithelium undergoes a progressive reduction in height, goblet cells and glands disappear, cartilage support is lost, and smooth muscle relative abundance peaks in the bronchioles to act as control valves for air distribution.

## Cell Types in Typical Respiratory Epithelium

▼ The typical respiratory epithelium is a pseudostratified ciliated columnar epithelium with goblet cells.

▼ All cells contact the basement membrane, but not all reach the apical surface, creating a pseudostratified appearance with rows of nuclei at different levels. Five cell types are present:

1. Ciliated Columnar Cells (approx. 30%): The most abundant type. Each cell has hundreds of robust cilia on its

apical surface that beat in unison to push mucus and trapped debris toward the pharynx (mucociliary escalator).

2. Goblet Cells (approx. 30%): Produce mucin glycoproteins to form the protective mucus layer. They appear pale or hollow in standard H&E sections. They decrease down the tree and are absent from terminal bronchioles

3. Basal (Stem) Cells (approx. 30%): Short, rounded cells resting on the thick basement membrane. They are

undifferentiated progenitor stem cells that undergo mitosis to regenerate all epithelial cell types.

4. Brush Cells (<10%): Columnar cells with apical microvilli and intracellular glycogen granules. They serve as sensory receptors specialized for the transduction of general chemical and tactile sensation.

5. Granule (Kulchitsky) Cells (<10%): Part of the diffuse neuroendocrine system. They contain dense core granules

of two types:

▼ catecholamine-secreting cells (neurosecretory) and

▼ protein-hormone secreting cells (enteroendocrine-like), which regulate local airway and vascular tone by releasing amines.

## SPECIALIZED UPPER RESPIRATORY STRUCTURES

▼ The Nasal Conchae (Turbinates): Three bony projections that break the air stream into turbulent eddies, greatly increasing the contact of air with the warm, moist mucosa. Particulate matter is thrown out of the stream (turbulent precipitation), adheres to the mucus, and is swept to the pharynx to be swallowed.

▼ Swell (Cavernous) Bodies: Specialized venous plexuses in the lamina propria of the conchae that resemble erectile

tissue. They engorge with blood periodically on alternate sides, temporarily reducing airflow on one side of the nasal cavity to allow its mucosa to recover from drying.

▼ Paranasal Sinuses: Air-filled cavities in the frontal, maxillary, ethmoid, and sphenoid bones. Lined by a thin respiratory epithelium with few goblet cells and a thin lamina propria continuous with the periosteum. Drainage

obstruction leads to sinusitis.

## The Olfactory Mucosa: Receptor Organ of Smell

Located in the roof of the nasal cavity, superior septum, and superior concha, the olfactory mucosa is thicker than the respiratory mucosa, has a distinctive yellowish-brown color, and lacks goblet cells. It consists of:

✓Olfactory Epithelium: A tall (100  $\mu\text{m}$ ) pseudostratified epithelium containing:

(1) Olfactory Cells: bipolar sensory neurons whose apical dendrites form olfactory knobs with non-motile cilia and basal axons form the unmyelinated olfactory nerve fibers.

(2) Sustentacular (Supporting) Cells: columnars with microvilli and yellow-brown lipofuscin granules. They express smooth ER, synthesize odorant-binding proteins (OBP), and act as glial-like metabolic and physical supports.

(3) Basal Cells: stem cells that continuously replace olfactory neurons (a rare example of adult neurogenesis).

(4) Brush Cells: general sensation transducers.

▼ Lamina Propria (Bowman's Glands): Pure serous, branched tubulo-alveolar glands. Their thin, watery secretion flushes the olfactory surface to dissolve odorants and wash away old stimuli, re-sensitizing the receptors.

## The Larynx and Vocal Folds

▼ Connects the pharynx with the trachea.

▼ The Epiglottis (elastic cartilage core) acts as a valve during swallowing.

▼ Its entire lingual surface and apical laryngeal surface are covered by protective nonkeratinized stratified squamous epithelium, which transitions to typical respiratory epithelium on its posterior laryngeal surface. The laryngeal cavity features two pairs of mucosal folds:

- Vestibular Folds (False Vocal Cords): Upper paired folds. Covered by respiratory epithelium. Contain seromucous glands, loose areolar tissue, and prominent MALT/lymphoid nodules. Do not play a role in phonation.

- Vocal Folds (True Vocal Cords): Lower paired folds projecting sharply into the lumen. Covered by protective

nonkeratinized stratified squamous epithelium. They contain the Vocal Ligament (dense band of elastic fibers)

and the Vocalis Muscle (skeletal muscle). Contraction adjusts tension and length, altering the gap (rima glottidis) to control pitch during vocalization.

## THE TRACHEOBRONCHIAL TREE

- ▼ Trachea: Formed of 16-20 U- or C-shaped hyaline cartilage rings joined posteriorly by the Trachealis (smooth muscle) against the esophagus. It is histologically identical to the primary bronchi.

- ▼ Intrapulmonary Bronchi: Secondary (lobar) and tertiary (segmental) bronchi branch within the lungs. They are

structurally distinguished from trachea/primary bronchi by:

(1) slightly folded mucosa

(2) the presence of a complete, spirally arranged muscularis mucosa smooth muscle layer that contracts post-mortem to cause mucosal folding, and

(3) discontinuous hyaline cartilage plates/patches rather than continuous rings.

## Terminal Bronchioles and Club (Clara) Cells

- ▼ Terminal bronchioles are the smallest air passages of the purely conducting portion. Their mucosa is lined by a ciliated simple columnar epithelium without goblet cells, supported by 1-2 layers of smooth

muscle cells. Interspersed in this epithelium are non-ciliated, dome-shaped exocrine secretory cells known as Club Cells (formerly Clara cells).

▼ Functions: They secrete a surfactant-like agent rich in phospholipids and lipoproteins, along with Club Cell Secretory Protein (CC16), a 16-kilodalton protein that alters the surface tension of the bronchiolar fluid layer to prevent airway collapse. Under toxic stress, Club cells act as stem cells to regenerate ciliated bronchiolar and

alveolar epithelial cells.

▼ Biomarker Significance: CC16 serves as a major pulmonary marker. During lung injury, Club cell damage

decreases CC16 levels in bronchoalveolar lavage fluid, while serum levels of CC16 increase because of leakage across the blood-air barrier.

▼ Nomenclature Note: Originally named 'Clara cells' by Dr. Max Clara in 1937 using lung tissue from executed

prisoners of the Nazi regime. Due to the ethical controversy, the term 'Club cell' was formally adopted on January 1, 2013.

## THE RESPIRATORY ZONE OF THE LUNG

▼ The respiratory zone represents the parenchymal lung tissue specialized for gas exchange. It begins at the

Respiratory Bronchioles, which are lined by ciliated simple cuboidal epithelium (containing both ciliated cells and Clara cells) and are characterized by walls interrupted by individual, saclike alveoli. At the alveolar junction, cuboidal epithelium transitions to simple squamous pneumocytes.

## Alveolar Epithelium and Cell Types

▼ Alveoli are cup-shaped blind pouches (up to 200  $\mu\text{m}$  in diameter). There are 150 to 250 million alveoli per adult lung, providing a massive 75  $\text{m}^2$  surface area (roughly a tennis court) for gas exchange.

▼ The alveolar wall consists of an extremely thin alveolar epithelium and three principal cell types:

1. Type I Pneumocytes (95% of Alveolar Surface Area): Extremely thin, flattened simple squamous epithelial cells. Although they line 95% of the surface area to minimize barrier thickness, they account for only about 40% of the pneumocyte cell population. They are highly sensitive to toxic injury and cannot divide.

2. Type II Pneumocytes (Septal Cells; 5% of Surface Area): Rounded or cuboidal secretory cells that make up 60% of the pneumocyte cell population. In H&E, they exhibit a foamy, vesicular peripheral cytoplasm due to membrane-bound lamellar bodies (1-2  $\mu\text{m}$ ) containing concentric phospholipid lamellae. They secrete pulmonary surfactant (decreases surface tension to prevent collapse during expiration) and act as the stem cells of the alveolar lining, dividing to regenerate both Type I and Type II pneumocytes following toxic gas inhalation or injury.

3. Alveolar Macrophages (Dust Cells): Active phagocytes located in the alveolar lumina and the interalveolar septa. They phagocytose airborne particulate matter (such as carbon/dust) and extravasated erythrocytes lost from damaged capillaries. They appear dark or brown due to carbon and hemosiderin. Most migrate up the mucociliary escalator to be swallowed in the pharynx; some exit via lymphatics or remain in septa for years.

#### The Blood-Air Barrier (Pulmonary Barrier)

▼ The minimal boundary separating alveolar air from capillary blood. It is highly optimized, measuring only 0.1 to 1.5  $\mu\text{m}$  in thickness. It is composed of three main layers:

1. Alveolar Epithelium (cytoplasm of Type I pneumocytes) and its thin basal lamina.

2. Interalveolar Connective Tissue Septum (the interstitium) consisting of elastic and reticular fibers, extracellular matrix, and stromal cells (fibroblasts, mast cells, leukocytes). It houses a dense network of continuous,

non-fenestrated blood capillaries.

3. Capillary Endothelium (cytoplasm of continuous endothelial cells) and its thin basal lamina.

Note on the Thinnest Portions: At sites of maximal gas exchange efficiency, the connective tissue septum is absent, and the basal laminae of the Type I pneumocyte and capillary endothelial cell are fused into a single membrane.

Comparison of Extra-pulmonary vs. Intra-pulmonary Bronchi